

FROZEN SITE-PROBE ACCURACY IS NOT A STANDALONE HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often judged with a site probe, a classifier that predicts the acquisition site: lower accuracy on harmonized images is read as success. We audit such verdicts, starting from a frozen probe trained on unharmonized images, on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites) with ten outputs of seven methods, one published (HACA3). Verdicts depended on probe training: five seeds of the frozen probe’s recipe gave the same CycleGAN output balanced accuracies from 0.13 to 0.59, and five longer-trained probes found the site more decodable than the frozen probe in eight of ten outputs. A Gaussian blur, moving intensities away from the target, scored lowest on the frozen probe, at the shuffled-label baseline, and HACA3’s preprocessing alone lowered frozen-probe accuracy from 0.95 to 0.61. Probes trained on harmonized images decoded the site label far better than raw-trained ones (+0.42 to +0.55 balanced accuracy on heads, +0.29 to +0.36 within the brain; chance 0.06), also from intensity histograms; HACA3 held raw-trained probes near 0.28, yet a histogram probe trained on its outputs reached 0.71. Low frozen-probe accuracy thus does not establish removal of site-label information. We recommend longer-trained and output-trained probes, a shuffled-label null, and target alignment apart from image change.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks on few scanners, so large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier that predicts the acquisition site from an image, which is easy on raw MRI [2]. If its accuracy

drops on harmonized images, site effects are taken to be removed [6, 7]; this is reported alongside similarity to the input, such as PSNR, and intensity-distribution distances. Some studies retrain the probe on harmonized images [6] or on image features [7, 8]. Others apply a probe trained on original images unchanged (a *frozen* probe): PRISM [9], for example, reports this drop, with target-site predictions and segmentation checks, as evidence of harmonization.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail because outputs are unfamiliar, damaged or noisy. Removing an attribute until a fixed adversary fails does not stop a new classifier from recovering it [10]. Site is decodable at about 0.9 chance-corrected balanced accuracy even from randomly initialized encoders [11]. Similarity to the input rewards doing nothing, and small change says nothing about the target.

Retrained classifiers are known to recover site after ComBat [12], histogram matching [13] and HACA3 [14]; how far probe protocols disagree on the same outputs has not been measured. Starting from the frozen probe of our earlier benchmark, on ten outputs of seven methods (one published, as released), we measure how the verdict depends on probe training, with controls for geometry, intensity, preprocessing and target alignment. We contribute the protocol, findings and a checklist (Sec. 5).¹ We do not rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [15] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented (in-plane rotation still varies), clipped to $[p_1, p_{99}]$ and scaled to $[0, 1]$; one axial slice per subject is chosen by foreground area, cropped to a dilated head mask and resized to 256×256 . NYU, the largest site, is the target; 19 test subjects are from NYU and 90 from 16 source sites. Slices are fixed and verified for every output. Labels are coarser than acquisitions: UM, UCLA and Leuven each

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

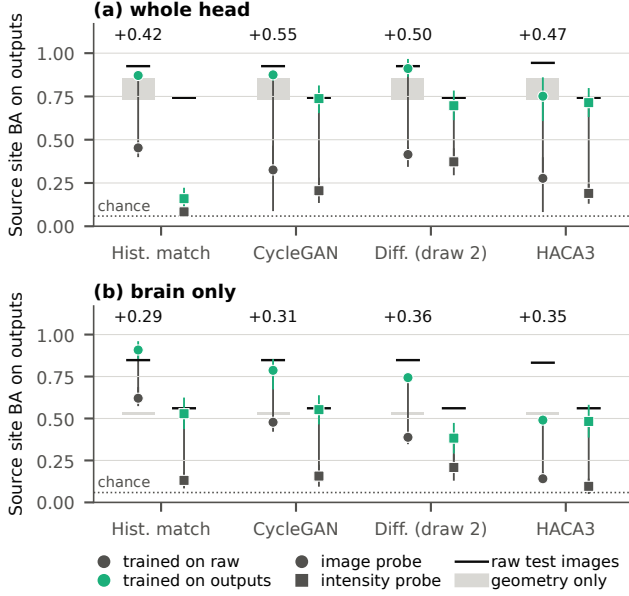


Fig. 1. Probes trained on raw slices (grey) or a method’s outputs (aqua), tested on its outputs (diffusion: draw 2), on (a) whole heads and (b) brains: image probes (circles, mean and range of 3 seeds) and histogram probes (squares, 95% interval). Numbers: paired image-probe difference. Black: raw-trained probe on raw slices. Grey: shape-only probes.

merge two released sub-samples, and 11 of 17 sites mix voxel sizes or matrices; no per-subject scanner identifier is released.

Audited outputs. Our earlier benchmark’s outputs were evaluated without retraining: NeuroCombat [1, 16] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a reference pooled over all 17 training sites, not NYU, as in the benchmark [17]; CycleGAN [18]; StarGAN [19], aggressive and conservative; a DLEST-style baseline without DLEST’s energy-based sampler [6], after 1,000 and 1,500 steps; conditional diffusion image-to-image translation [20, 21] (strength 0.35, 50 sampling steps), sampled twice from one checkpoint. HACA3 [22] was used as released (pretrained weights, NYU target image) after its required N4 correction and rigid MNI registration, mapped back to native space: ten outputs of seven methods (Table 1 omits DLEST-style 1500, conservative StarGAN). As a control ignoring the target, raw test slices were blurred (Gaussian, $\sigma = 2, 4, 8$ pixels). Other learned methods used the training split. Test subjects were seen in the earlier benchmark (retrospective reanalysis); each analysis was fixed in a dated protocol before its results.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output

with input on whole images; most target-conditioned generators return NYU inputs almost unchanged, so we summarize the 90 source subjects. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities (defined on the raw slice) and a reference pooled from all 147 NYU training subjects; ΔW_{NYU} is the output’s distance minus the input’s.

Site probes. Image probes are single-channel ResNet-18s [23] predicting 17 sites, tested on the fixed slices; unless stated otherwise they use AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation and are selected on validation balanced accuracy (BA). We use five probe families:

Frozen (1): the earlier benchmark’s own checkpoint.

Benchmark recipe (5 seeds): the frozen probe’s recipe with new seeds, a random raw-volume axial slice per step.

Converged (5 seeds): four times the steps, cosine decay, reseeded data-loader workers, final epoch.

Shuffled labels (4): labels permuted between subjects.

Slice probes (3 seeds per input, plus restricted networks and logistic fits): trained on one fixed slice per training subject, raw or one method’s outputs. Restricted variants see only silhouettes or brain masks (geometry), intensity histograms (logistic regression) or HD-BET brain-only slices [24].

Slice-probe methods were set in advance (intensity, adversarial, diffusion); converged, silhouette, histogram and brain-only probes came after earlier results but before their own.

Statistics. Source BA averages the 16 source sites (2–15 test subjects each). Uniform guessing over 17 classes scores $1/17 \approx 0.06$, one source class for every subject $1/16$; shuffled-label probes (empirical null) scored 0.03–0.08. Intervals are 95% percentiles of 2,000 within-site bootstrap replicates, paired across methods and probes; they condition on these sites and checkpoints, not retraining or cohorts.

4. RESULTS

Probe validity. Raw source BA was 0.95 for the frozen probe, 0.81–0.98 for benchmark-recipe and 1.00 for converged probes. Probes learn more than scanner appearance: head silhouettes alone gave 0.73–0.86, and age, sex and diagnosis 0.22 [0.15, 0.29] (0.31 with brain volume).

Verdicts depend on probe training. Benchmark-recipe probes gave very different site BA to the same outputs (Table 1): CycleGAN 0.13–0.59 and aggressive StarGAN 0.18–0.54, Kendall τ 0.39–0.83 between seeds’ method rankings (nine outputs). Raw accuracy did not flag it: a probe with raw BA 0.98 gave CycleGAN 0.20. A longer, stabilized recipe removed much of this spread (validation BA within 0.035 over the last five epochs; τ 0.78–0.94; per-output ranges ≤ 0.17); as it changes several factors, it shows recipe sensitivity, not its cause. Every converged probe exceeded the frozen probe’s

Table 1. Source-subject means ($n = 90$); arrows: direction usually read as better. PSNR, XCorr: similarity to the input; W : input–output distance; ΔW_{NYU} , KL_{NYU} (output || NYU): distance to NYU. Frozen probe: BA [95% CI], $\rightarrow\text{NYU}$: % of outputs it labels NYU; benchmark recipe (the frozen probe’s, new seeds), converged: mean (min–max) of 5 seeds; blur: frozen probe only. [†]Transductive. [‡]Pooled 17-site reference, not NYU. [§]KL high as CSF is near zero.

Method	Change			Target alignment		Site BA: probes trained on unharmonized images, tested on each row			
	PSNR $\uparrow$	XCorr $\uparrow$	W	$\Delta W_{\text{NYU}}\downarrow$	$\text{KL}_{\text{NYU}}\downarrow$	Frozen probe $\downarrow$	$\rightarrow\text{NYU}\uparrow$	Benchmark recipe $\downarrow$	Converged $\downarrow$
Raw input	–	–	0	0	0.30	0.95	0%	0.90 (0.81–0.98)	1.00 (1.00–1.00)
NeuroCombat [†]	20.0	0.940	0.055	+0.003	0.88	0.27 [0.21, 0.32]	2%	0.33 (0.26–0.37)	0.34 (0.28–0.42)
Histogram matching [‡]	23.0	0.982	0.054	-0.053	0.09	0.44 [0.37, 0.52]	10%	0.44 (0.27–0.59)	0.61 (0.55–0.68)
CycleGAN	21.3	0.977	0.048	-0.036	0.25	0.31 [0.25, 0.37]	52%	0.28 (0.13–0.59)	0.49 (0.44–0.54)
StarGAN (aggr.)	16.0	0.896	0.071	-0.019	0.32	0.21 [0.15, 0.27]	56%	0.35 (0.18–0.54)	0.70 (0.65–0.81)
DLEST-style 1000	28.4	0.993	0.017	-0.009	0.31	0.71 [0.65, 0.77]	6%	0.84 (0.76–0.89)	0.99 (0.96–1.00)
Diffusion, draw 1	21.7	0.948	0.008	-0.001	0.83	0.37 [0.32, 0.42]	36%	0.42 (0.24–0.63)	0.57 (0.52–0.59)
Diffusion, draw 2	21.7	0.948	0.008	-0.002	0.82	0.35 [0.29, 0.40]	38%	0.42 (0.30–0.62)	0.56 (0.50–0.59)
HACA3 (pretrained) [§]	18.9	0.954	0.062	-0.016	2.29	0.25 [0.18, 0.32]	40%	0.29 (0.18–0.37)	0.28 (0.23–0.33)
Blur control, $\sigma=2$	24.7	0.974	0.018	+0.016	0.57	0.07 [0.06, 0.10]	0%	–	–

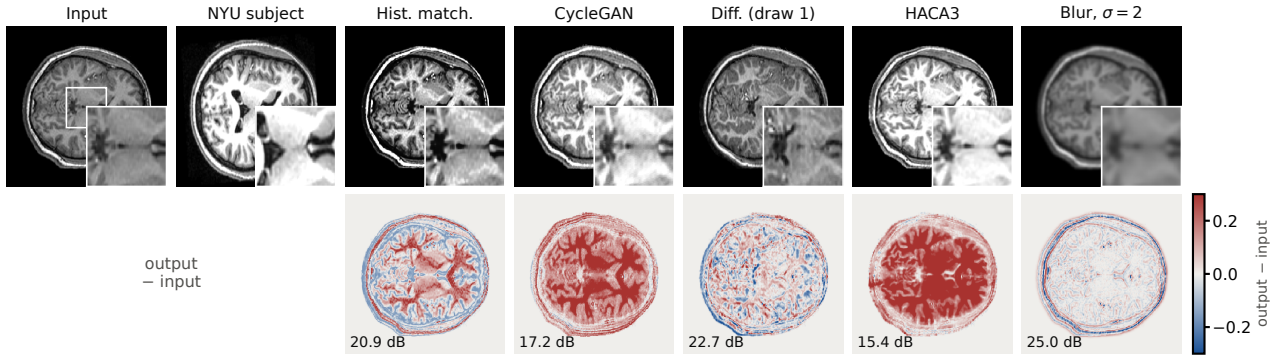


Fig. 2. A median-PSNR source subject and an axial NYU slice with similar head outline and anatomy, in the input’s window; insets: 2 $\times$ zoom of the box. Bottom: output – input with PSNR. Blur has the highest PSNR shown.

interval on all outputs except NeuroCombat and HACA3, e.g. aggressive StarGAN 0.65–0.81 vs. 0.21.

A blur scored lowest on the frozen probe. A $\sigma = 2$ blur (Fig. 2) kept PSNR at 24.7 dB, above CycleGAN and diffusion, and moved intensities away from NYU ($\Delta W_{\text{NYU}} = +0.016$), yet scored frozen-probe BA 0.07 [0.06, 0.10], in the shuffled-label range ($\sigma = 4, 8$: 0.06, 0.07). It labelled 87 of 90 blurred slices Yale, none NYU; on CycleGAN, aggressive StarGAN and diffusion 65–80% of errors named NYU, so target-site predictions (as PRISM checks) separate them, but miss retained site information (HACA3, below). A target-blind perturbation can thus collapse frozen-probe accuracy.

Is site still decodable? Slice probes trained on raw slices reached BA 0.92 on raw test slices but 0.45, 0.33 and 0.41 on histogram-matched, CycleGAN and diffusion outputs, a drop that, taken alone, reads as harmonization [9]. Probes trained on each method’s outputs recovered the site at 0.87, 0.87 and 0.91, +0.42 [0.36, 0.48], +0.55 [0.52, 0.58] and +0.50 [0.45, 0.54] over raw-trained probes (Fig. 1a; paired three-seed means; the benchmark recipe may understate decodability). As head geometry identifies sites, we repeated this within the brain: output-trained probes still exceeded raw-trained ones

by +0.29 to +0.36 (Fig. 1b; all intervals above zero, also with final-epoch checkpoints), and brain masks alone gave 0.52–0.54. Intensity histograms (squares) showed the same gap within the brain (output-trained 0.38–0.55, raw-trained 0.13–0.21); on whole heads only histogram matching, matched by construction, hid the site (0.16). Low frozen-probe accuracy thus does not show removal of site-label information.

A published harmonizer. HACA3 kept every raw-trained probe low (frozen 0.25 [0.18, 0.32], benchmark recipe 0.18–0.37, converged 0.23–0.33, slice probes 0.28; 54% of frozen-probe errors named NYU), yet probes trained on its outputs recovered the source site at 0.75 (+0.47 [0.43, 0.52]) and within the brain at 0.49 (+0.35 [0.30, 0.40]; Fig. 1), below brain masks alone (0.52–0.54); intensity histograms, which see no shape, gave 0.71 [0.63, 0.80] on heads and 0.48 within the brain (raw-trained 0.19, 0.10). Its preprocessing alone (N4, registration, mapping back) gave frozen BA 0.61 [0.53, 0.68], 24% of errors naming NYU; HACA3 lowered it by a further 0.35 [0.26, 0.44]. Five-fold cross-validation on all 928 source subjects (site-stratified folds, 10% of training data for model selection, held-out predictions pooled) agreed: raw-trained probes 0.25 on HACA3, HACA3-trained 0.77

(+0.52 [0.48, 0.56]).

Small change is not target alignment. Diffusion changed the intensity distribution least ($W = 0.008$) yet barely moved it toward NYU ($\Delta W_{\text{NYU}} = -0.0013$, 1% of the raw distance) and raised KL to NYU from 0.30 to 0.83. A second diffusion draw differed from the first more than from the input (median absolute difference 0.085 vs. 0.075).

5. DISCUSSION AND RECOMMENDATIONS

One site number is not a verdict: it moved by up to 0.46 across seeds, was lowest for a blur, and missed the site-label information in HACA3’s intensity histograms. We recommend (1) several converged probes, their recipe, raw accuracy, a shuffled-label null and similarity to the input; (2) output-trained probes with channel-restricted and geometry controls, whose high accuracy shows retained site-label information, not by itself failed harmonization; (3) intensity alignment to the target apart from change: the least-changed output closed 1% of its distance to NYU; (4) paired uncertainty over subjects and draws.

Limitations. We use one cohort, one target site and one axial slice per subject (no volumetric effects), with 2–15 test subjects per source site. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Site is confounded with demographics, brain shape, slice level and acquisition geometry; our controls bound these channels but cannot separate scanner from biological effects. Except HACA3, methods are our untuned implementations, one transductive: a focused counterexample, not a survey.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [15], whose sites shared the fully anonymized data with local ethics approval or an explicit waiver. Ethical approval was not required as confirmed by the license attached with the open access data.

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